

Systems & Product Design Portfolio

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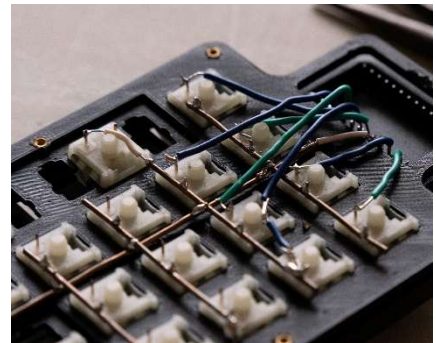


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Industrial / Human Factors Design

Projects that demonstrate systems design thinking, embedded systems, physical prototypes, human-centered design, and manufacturing techniques.

Minecraft Companion Device

Embedded electronics and systems, real-time game integration, UI/UX design, human factors design, and media production.

Audio Spectrum Visualizer Desktop Device

Embedded electronics and systems, Windows integration, CAD, modular design

Printable Arc Notebook

CAD, Modular design, DFM

Minecraft Companion Device

<https://www.youtube.com/watch?v=wyKZOjsitpY>

Overview

- A Minecraft game companion that displays **live game data**, extracted using a java-based game modification, and displayed on a 2" LCD screen using a sprite sheet
- Application of **Don Norman's** principles of design to the physical controls and UI design, including heuristics like discoverability and feedback.
- A fully 3D printed project enclosure with PLA and PETG
- Modelled in **SolidWorks**, coded in **C++** (Arduino IDE), **Python** 3.11.9, and JVM Gradle 9.5.1

Implementation

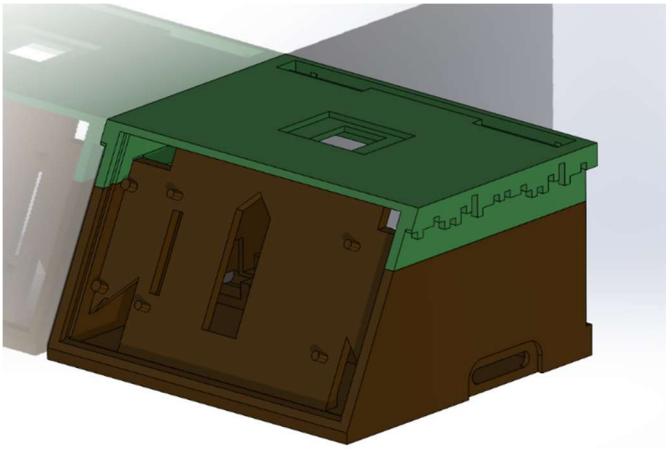
- A companion device that extracts live player telemetry data from an external software environment through a java based game mod, using an **ESP32** microcontroller to display the player telemetry through a sprite based UI and two individually addressed LED strips.
- The first mode contains navigation-based information including a compass that points to the player's spawn point which is an in-game static 3D point set by the player.



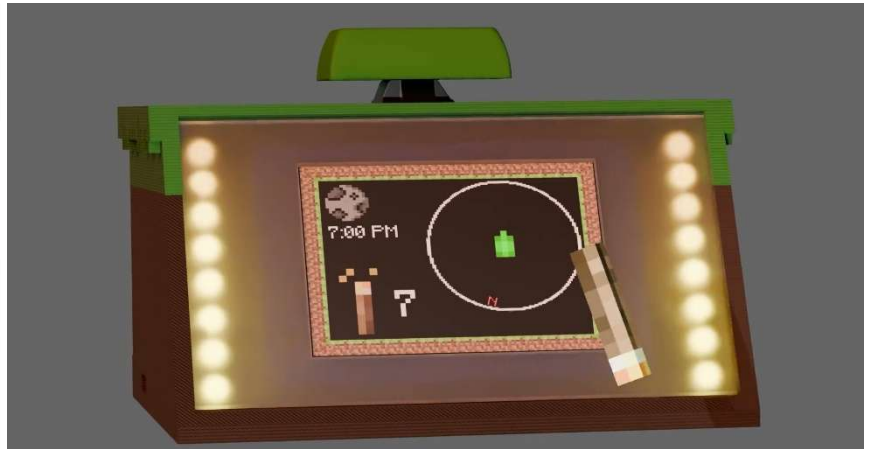
- The second mode contains combat-based information including tracking nearby hostile enemies, prioritizing those that are close to the player.
- A full **YouTube video production** including filming, a Blender animation, Minecraft animation, and motion graphics.

Next Steps

- Explore advanced manufacturing techniques such as including **laser cutting**, **forming**, and **CNC** (wood) for improved manufacturability and product feel.
- Apply **5S LEAN** methodologies to improve assembly organization, and repeatability.
- Reduce enclosure volume through PCB integration and component miniaturization.



Final Solidworks Assembly for fit



Blender render showcasing Mode 2



Minecraft scene showing Discoverability Heuristic



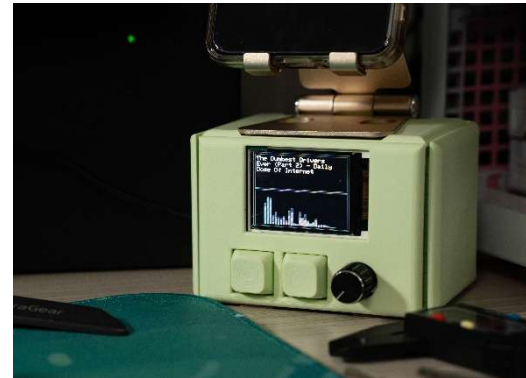
Intro of YouTube video

Audio Spectrum Visualizer Desktop Device

<https://www.youtube.com/watch?v=QmaiqrSgZdo>

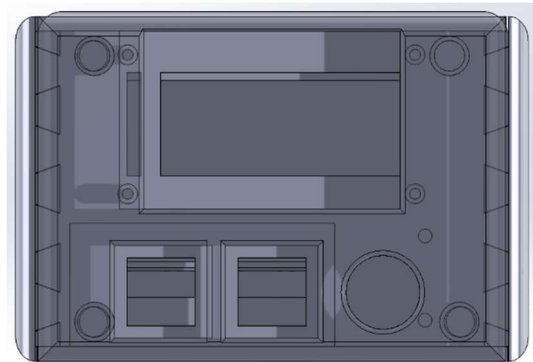
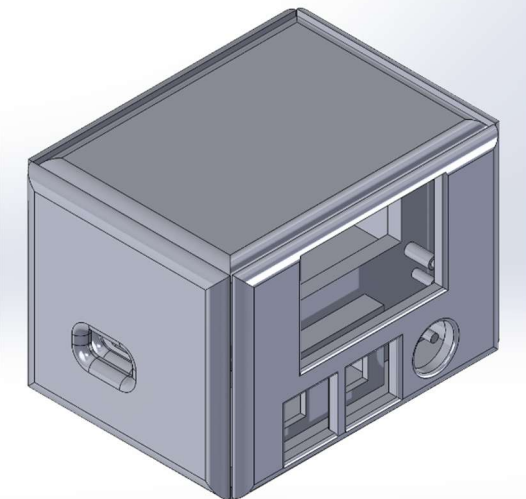
Overview

- A desktop device which analyzes Windows media playback and displays a real-time audio histogram alongside current media metadata.
- Primarily 3D-printed (as it's supposed to be replicable and user-modular) using PLA for all structural components with PETG and copper foil tape for the 3D printed PCB. (**Bambu Lab P1S**)
- Bottom / Top mount made of wood, milled using the **Roland MDX-40 CNC Mill**.
- Modelled in **SolidWorks**, coded in **C++** (Arduino IDE) and **Python** 3.11.9.



Implementation

- The device performs spectral analysis on Windows media playback, sends band audio data to an **ESP32** controller via serial communications, and displays it on a 2-inch TFT display along with media metadata.
- The device can change Windows volume, play/pause and skip/back media with the buttons and rotary encoder via **quadrature decoding** and filtering for the rotary encoder, and serial communications.
- Magnetic side panels in combination with dovetail joints for the main frame create a modular and structurally sound construction that can be customized.

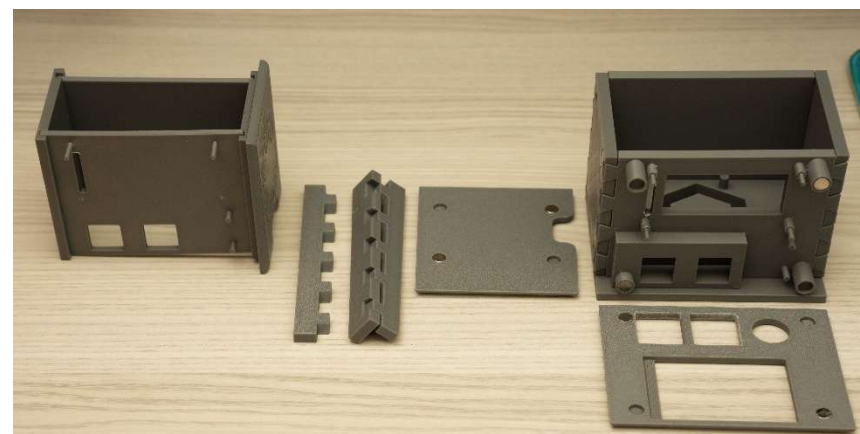


Next Steps

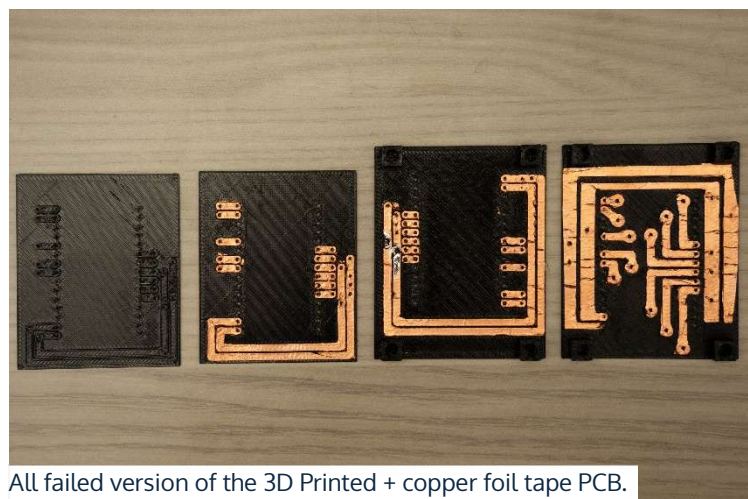
- Apply 5S LEAN methodologies to create a DIY kit of the device that can be repeated and created with minimal Engineering experience.
- Create the product trailer video with media captured from the design and production process and create a suite of mock advertising media for the product.
- Explore manufacturing methods by creating a wooden knob, keycaps, and shrouds as replacements for the 3D printed components. (Epilog Fusion M2 Laser Cutter or more CNC milled components)



Early Prototype of the device. All electrical components were fully functional.



Version 1 pictured on the left, Dovetail joint tests in the middle, and Version 2 pictured on the right



All failed version of the 3D Printed + copper foil tape PCB.

Printable Arc Notebook

- A reverse-engineered system to emulate the existing *Staples Arc Notebook* system, a modular binder/notebook system
- Designed to be fully 3d printable which also heavily restricted material use and form
- The main notebook components were engineered to fit with already existing paper holepunch tolerances
- A mix of materials and properties were explored in this project (PLA, TPU95A)

Implementation

- A working emulation that is easily printable, assemble-able, and opens an entire ecosystem of printable accessories
- Iterative designs to approach a design that is polished, works like the native system, and feels satisfying to use
 1. Early versions did not feel good to use because of the thickness and geometry of the covers, the carry size, and "flipability" were concerns)
- Wear testing was done on the circular spine rings and the covers to test the feasibility of the final design printed in one singular material
 1. Repeated insertions of the cover to the circular spines as well as bending the cover just to the point of plastic deformation (figures
- Iterations included remeasuring and reprinting to get tolerances correct and snug
- Experimentation of material properties (PLA and TPU95A), as early versions of the cover included a TPU95A spine



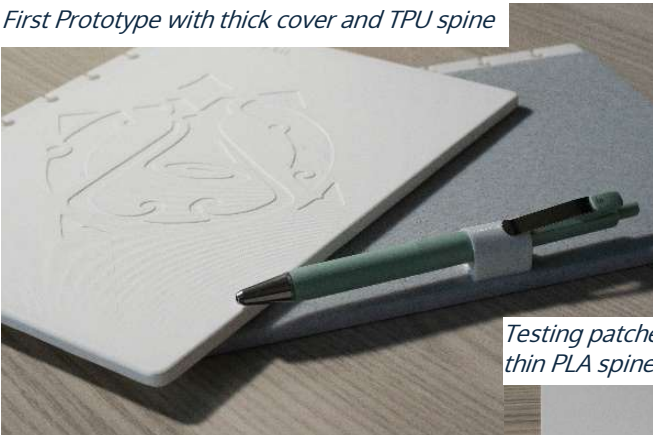
Final prototype onto of previous prototypes



First Prototype with thick cover and TPU spine



First Prototype with thick cover and TPU spine



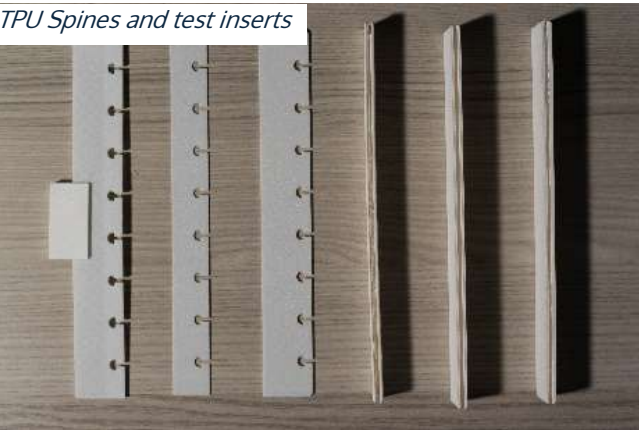
Testing patches for the thin PLA spine



Testing patches for the thin PLA spine



TPU Spines and test inserts



TPU Spine



TPU Pen holder variations for tolerances

